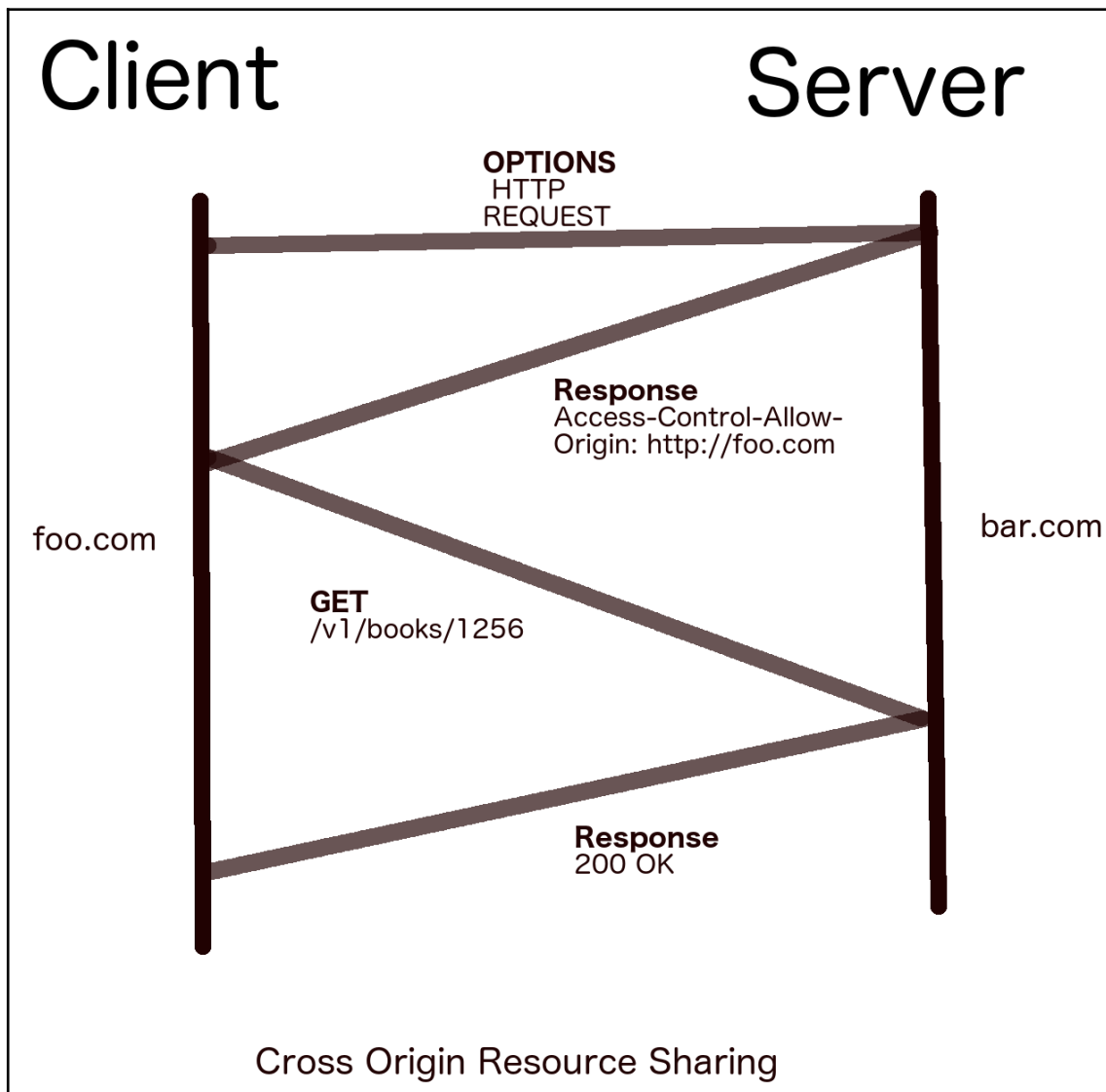
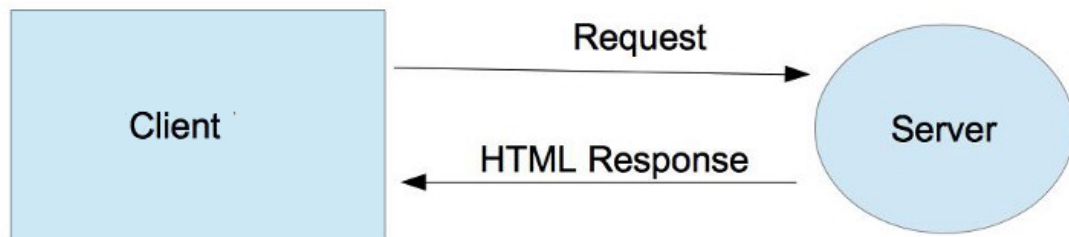


Chapter 1: Getting Started with REST API Development



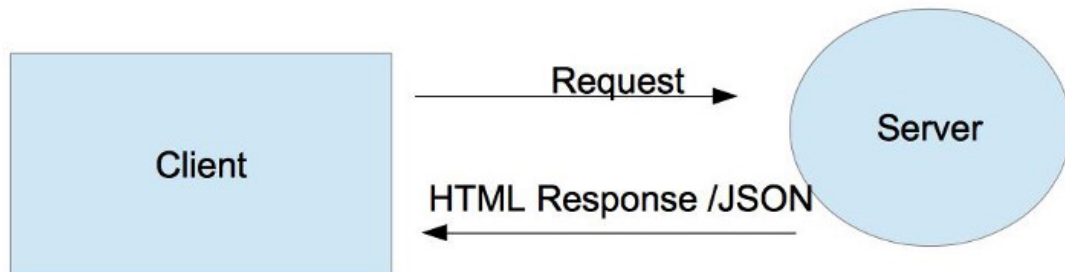
Old Way



- All the time



New Way



- HTML Response(first time)

- JSON from there on..



```
bin/  
  hello          # command executable  
pkg/  
  linux_amd64/   # this will reflect your OS and architecture  
    github.com/user/  
      stringutil.a # package object  
src/  
  github.com/user/  
    hello/  
      hello.go    # command source  
    stringutil/  
      reverse.go  # package source
```

Mirror Finder Service 0.1.1 OAS3

[./openapi.json](#)

API service for finding the fastest mirror from the list of given mirror sites

Servers

[http://localhost:8000](#) - Development server[Staging/Production are different from this] ▾

default ▾

GET **/fastest-mirror** Returns a fastest mirror site.

This call returns data of fastest reachable mirror site

Parameters

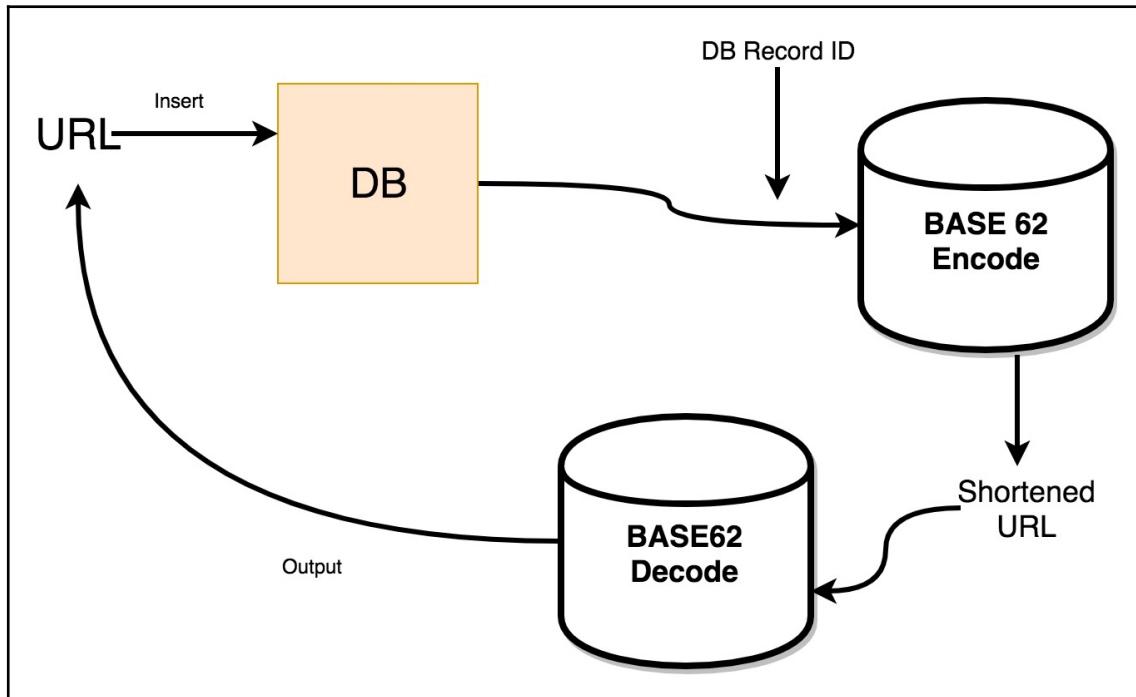
[Try it out](#)

No parameters

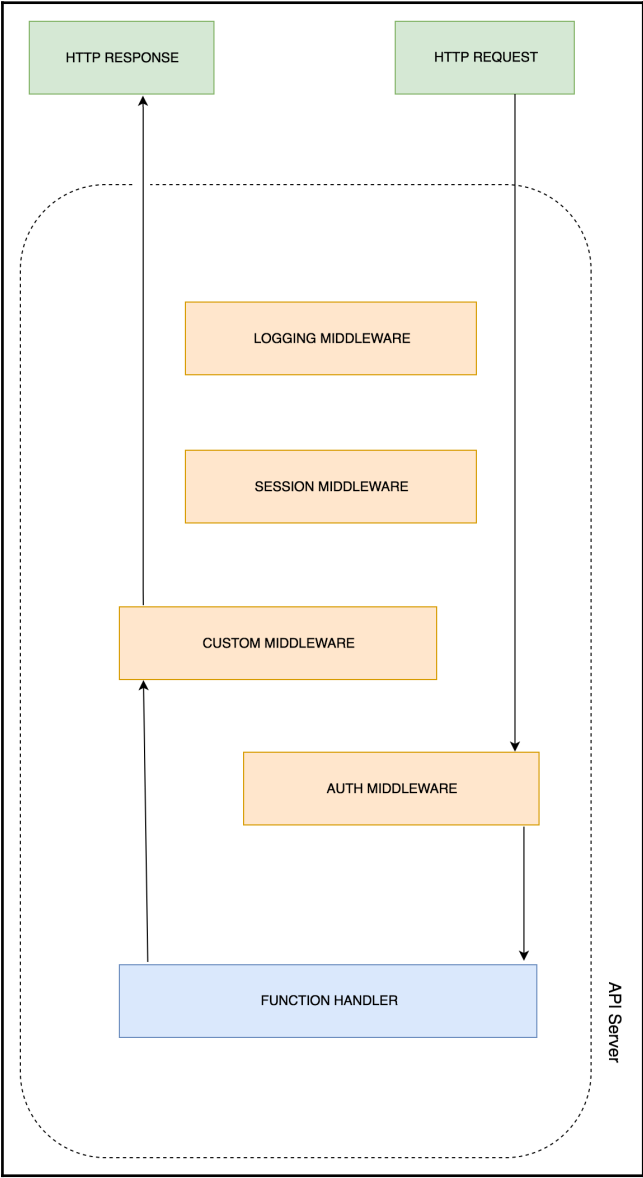
Responses

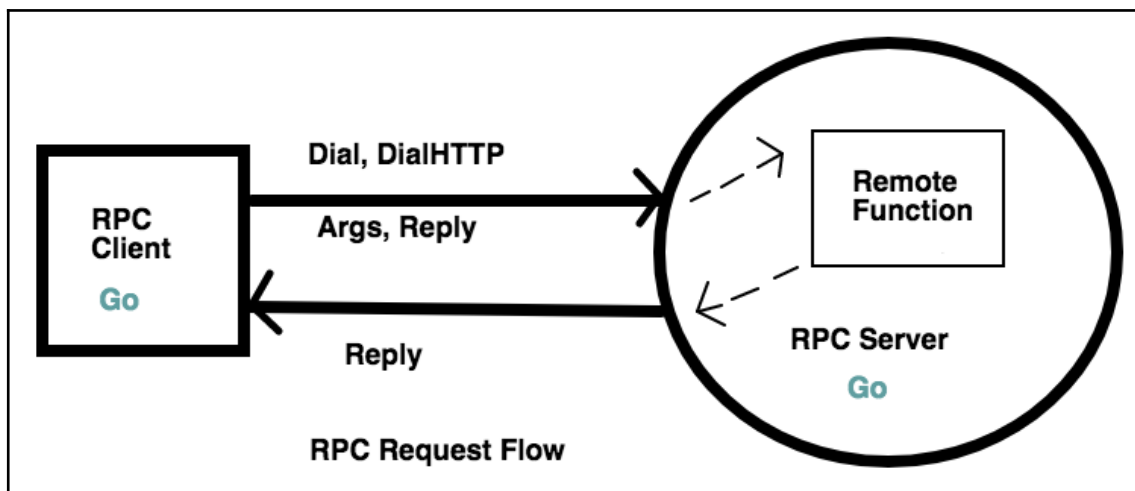
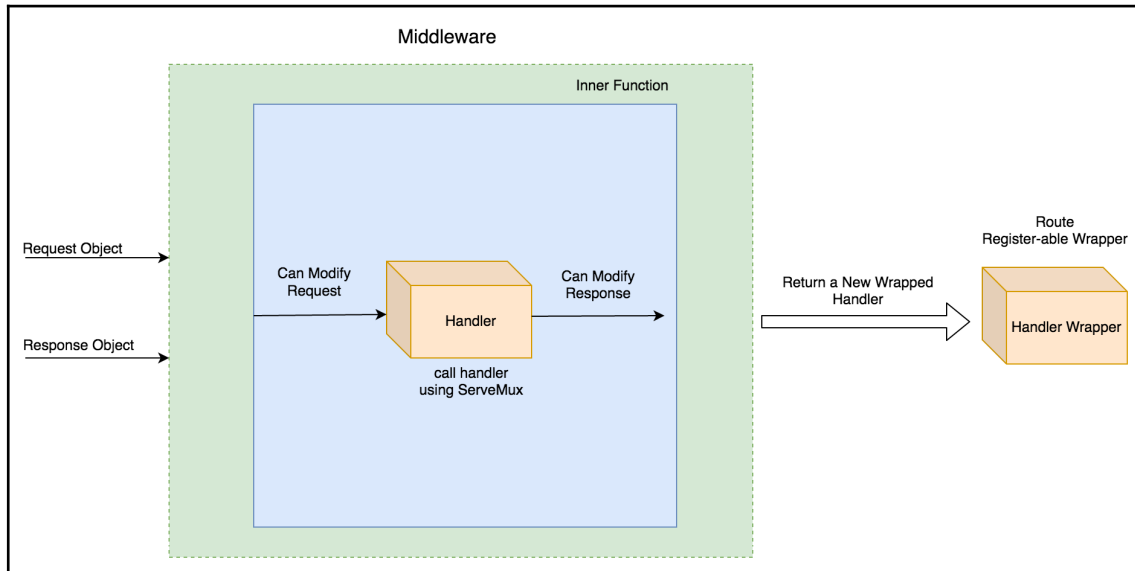
Code	Description	Links
200	A JSON object of details application/json ▾ Controls Accept header. Example Value Schema <pre>{ "fastest_mirror": "string", "latency": 0}</pre>	No links

Chapter 2: Handling Routing for our REST Services



Chapter 3: Working with Middleware and RPC





Chapter 4: Simplifying RESTful Services with Popular Go Frameworks

```
Hands-On-RESTful-Web-services-with-Go/chapter4 on □ develop [!]
> go run ginExample/main.go
[GIN-debug] [WARNING] Creating an Engine instance with the Logger and Recovery middleware already attached.

[GIN-debug] [WARNING] Running in "debug" mode. Switch to "release" mode in production.
- using env:   export GIN_MODE=release
- using code:  gin.SetMode(gin.ReleaseMode)

[GIN-debug] GET    /pingTime      --> main.main.func1 (3 handlers)
[GIN-debug] Listening and serving HTTP on :8000
[GIN] 2020/02/27 - 20:08:05 | 200 |      152.716µs |      ::1 | GET /pingTime
```

```
Hands-On-RESTful-Web-services-with-Go/chapter4 on □ develop [!] via 85 v1.13.5
> go run railAPIGin/main.go
All tables created/initialized successfully!
[GIN-debug] [WARNING] Creating an Engine instance with the Logger and Recovery middleware already attached.

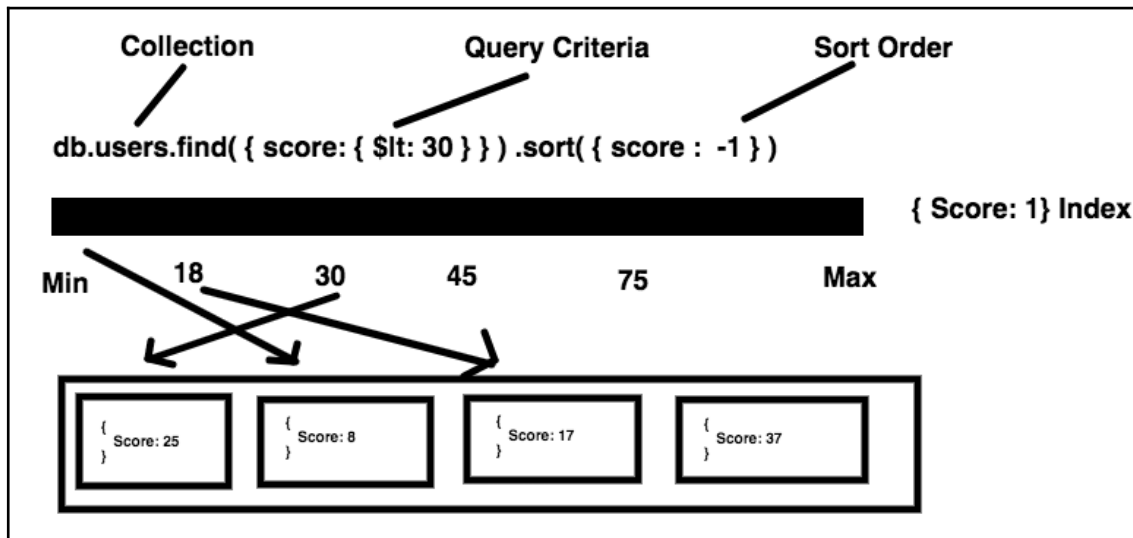
[GIN-debug] [WARNING] Running in "debug" mode. Switch to "release" mode in production.
- using env:   export GIN_MODE=release
- using code:  gin.SetMode(gin.ReleaseMode)

[GIN-debug] GET    /v1/stations/:station_id --> main.GetStation (3 handlers)
[GIN-debug] POST   /v1/stations             --> main.CreateStation (3 handlers)
[GIN-debug] DELETE /v1/stations/:station_id --> main.RemoveStation (3 handlers)
[GIN-debug] Listening and serving HTTP on :8000
[GIN] 2020/02/04 - 08:38:01 | 200 |      3.31529ms |      ::1 | POST /v1/stations
[GIN] 2020/02/04 - 08:38:22 | 200 |      593.308µs |      ::1 | GET /v1/stations/1
[GIN] 2020/02/04 - 08:38:28 | 200 |      250.335µs |      ::1 | DELETE /v1/stations/1
```

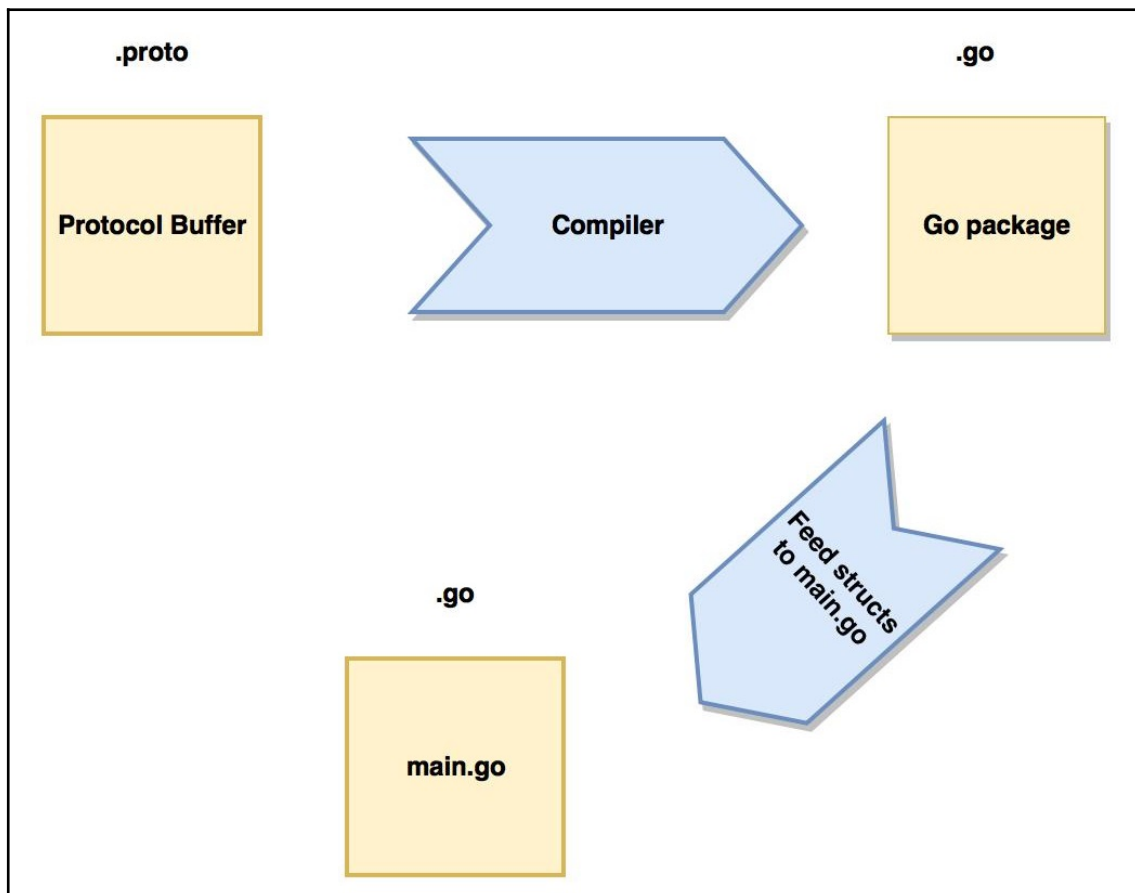
Chapter 5: Working with MongoDB and Go to Create a REST API

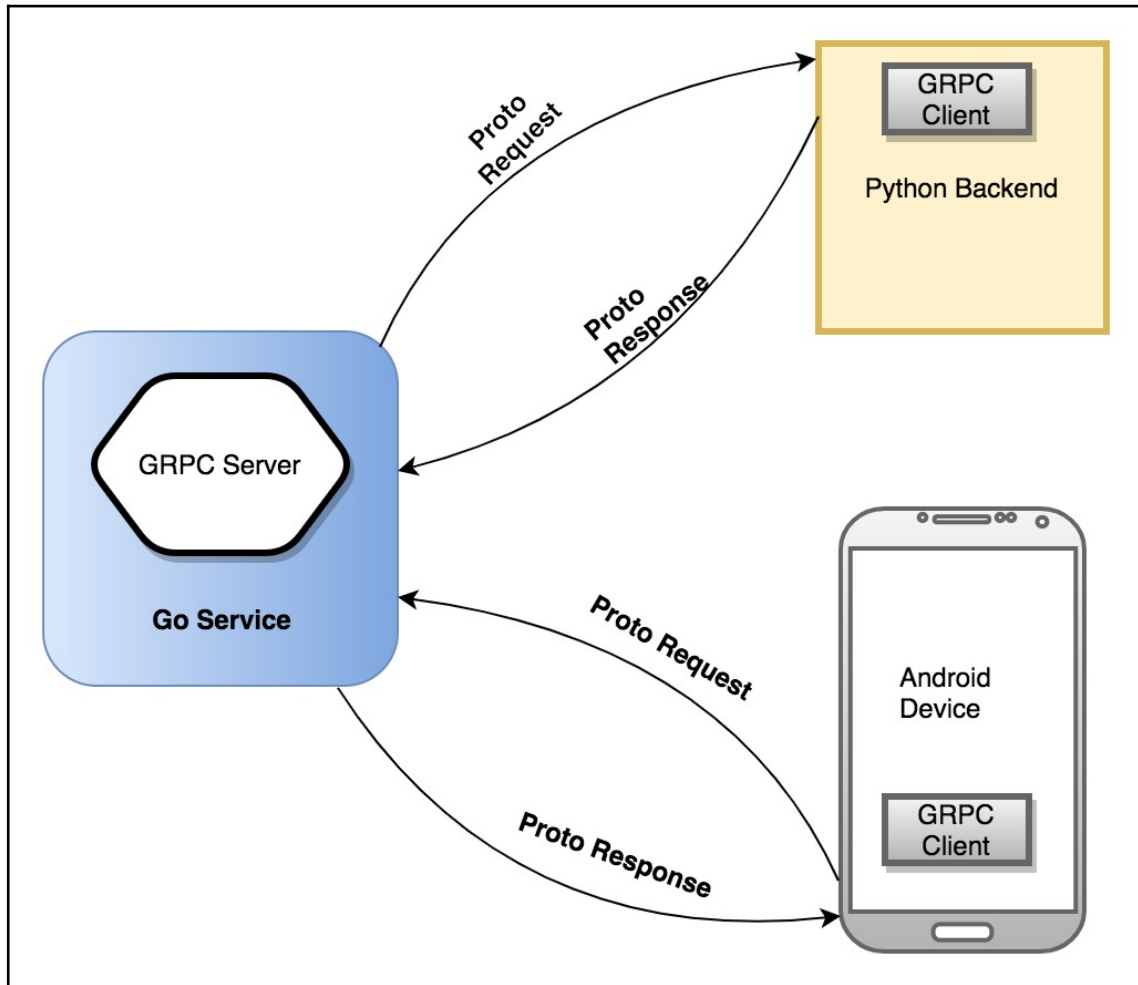
```
> mongod
2020-02-09T10:42:47.171+0100 I CONTROL [main] Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] MongoDB starting : pid=10615 port=27017 dbpath=/data/db 64-bit host=naren.local
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] db version v4.2.3
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] git version: 6874650b362138df74be53d366bbefc321ea32d4
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] allocator: system
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] modules: none
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] build environment:
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] distarch: x86_64
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] target-arch: x86_64
2020-02-09T10:42:47.193+0100 I CONTROL [initandlisten] options: {}
2020-02-09T10:42:47.194+0100 I STORAGE [initandlisten] wiredtiger-open config: create,cache_size=3584M,cache_overflow=(file_max=0M),session_max=33000,eviction=(threads_min
=4,threads_max=4),config_base=false,statistics=(fast),log=(enabled=true,archive=true,path=journal,compressor=snappy),file_manager=(close_idle_time=100000,close_scan_interval
=10,close_handle_minimum=250),statistics_log=(wait=0),verbose=[recovery_progress,checkpoint_progress],
2020-02-09T10:42:48.051+0100 I STORAGE [initandlisten] WiredTiger message [1581241368:51170][10615:0x7ffffa57bc380], txn-recover: Set global recovery timestamp: (0, 0)
2020-02-09T10:42:48.220+0100 I RECOVERY [initandlisten] WiredTiger recoveryTimestamp. Ts: Timestamp(0, 0)
2020-02-09T10:42:48.305+0100 I STORAGE [initandlisten] Timestamp monitor starting
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten]
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] ** WARNING: Access control is not enabled for the database.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Read and write access to data and configuration is unrestricted.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten]
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] ** WARNING: This server is bound to localhost.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Remote systems will be unable to connect to this server.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Start the server with --bind_ip <address> to specify which IP
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      addresses it should serve responses from, or with --bind_ip_all to
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      bind to all interfaces. If this behavior is desired, start the
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      server with --bind_ip 127.0.0.1 to disable this warning.
```

```
> mongo
MongoDB shell version v4.2.3
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("6cf90f5a-5a8a-46bb-b7b8-378f6f2cbe45") }
MongoDB server version: 4.2.3
Server has startup warnings:
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten]
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] ** WARNING: Access control is not enabled for the database.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Read and write access to data and configuration is unrestricted.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten]
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] ** WARNING: This server is bound to localhost.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Remote systems will be unable to connect to this server.
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      Start the server with --bind_ip <address> to specify which IP
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      addresses it should serve responses from, or with --bind_ip_all to
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      bind to all interfaces. If this behavior is desired, start the
2020-02-09T10:42:48.308+0100 I CONTROL [initandlisten] **      server with --bind_ip 127.0.0.1 to disable this warning.
---
```



Chapter 6: Working with Protocol Buffers and gRPC





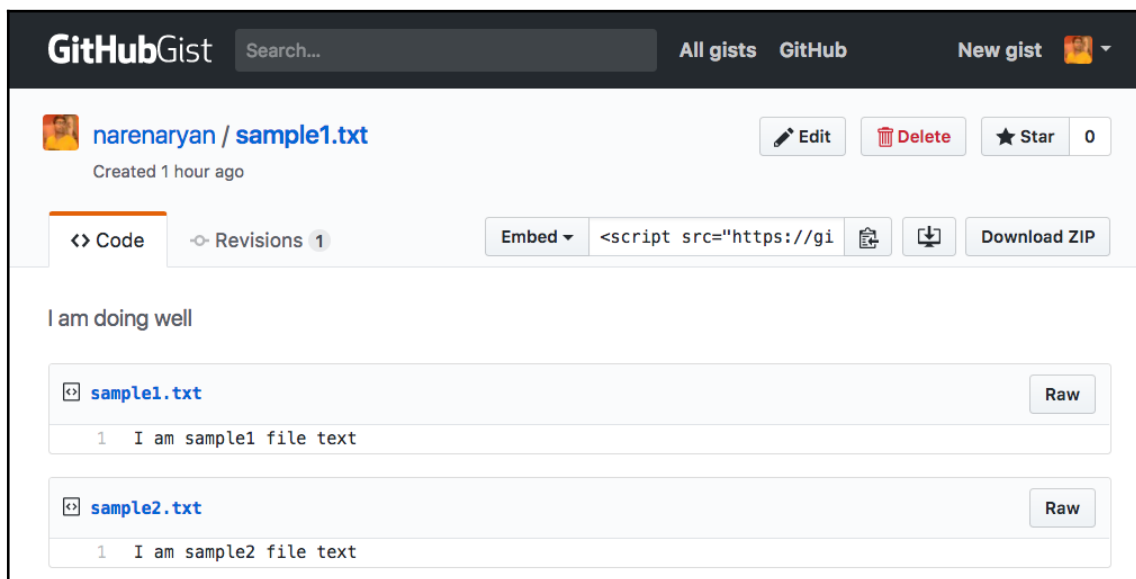
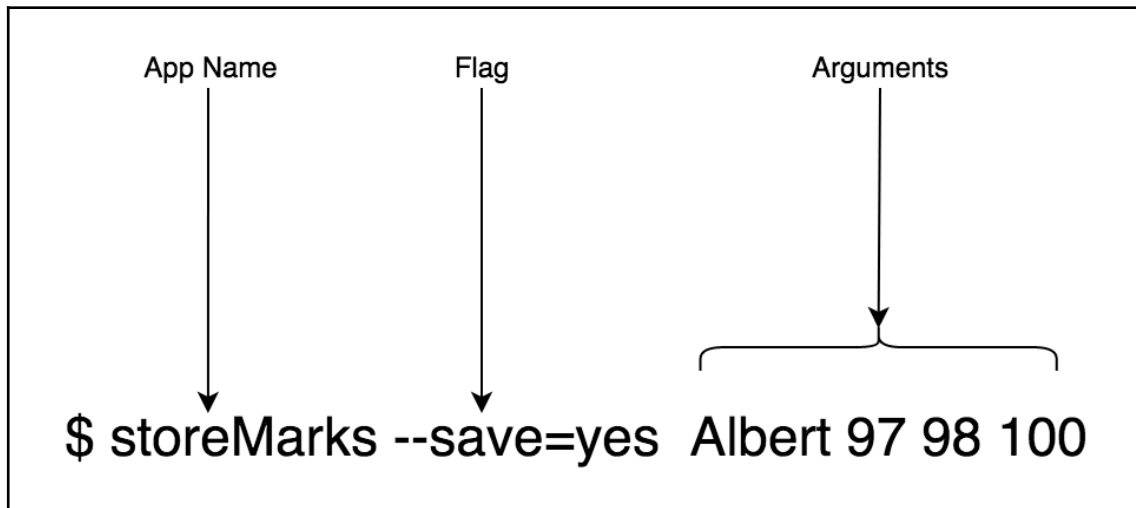
Chapter 7: Working with PostgreSQL, JSON, and Go

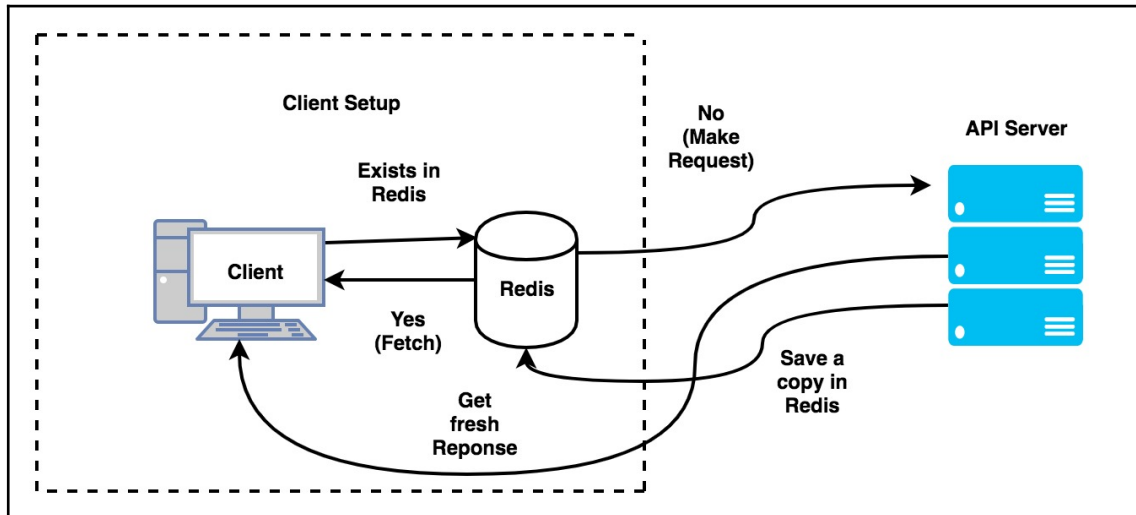
Informational

(options: S = show system objects, + = additional detail)

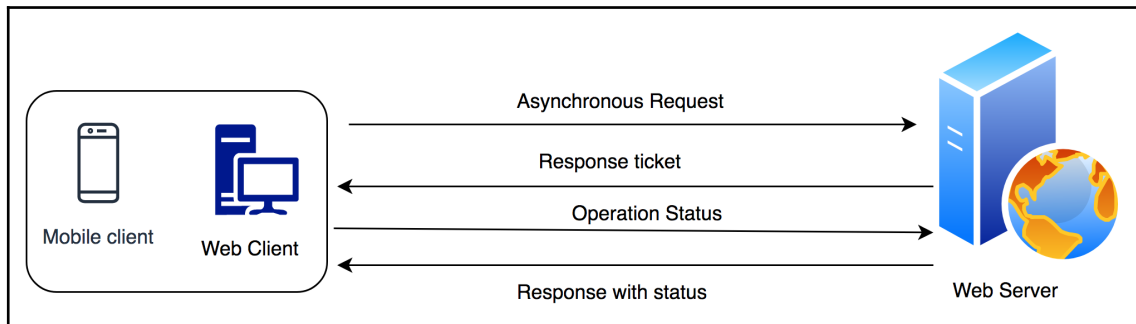
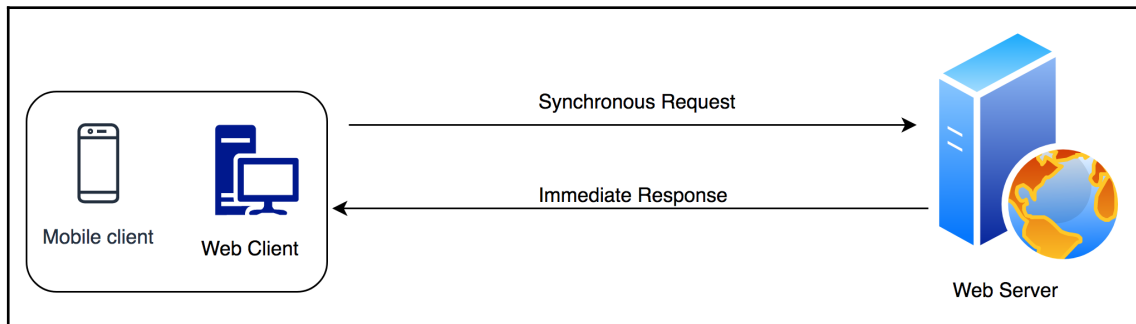
<code>\d[S+]</code>		list tables, views, and sequences
<code>\d[S+] NAME</code>		describe table, view, sequence, or index
<code>\da[S] [PATTERN]</code>		list aggregates
<code>\dA[+] [PATTERN]</code>		list access methods
<code>\db[+] [PATTERN]</code>		list tablespaces
<code>\dc[S+] [PATTERN]</code>		list conversions
<code>\dC[+] [PATTERN]</code>		list casts
<code>\dd[S] [PATTERN]</code>		show object descriptions not displayed elsewhere
<code>\ddp [PATTERN]</code>		list default privileges
<code>\dD[S+] [PATTERN]</code>		list domains
<code>\det[+] [PATTERN]</code>		list foreign tables
<code>\des[+] [PATTERN]</code>		list foreign servers
<code>\deu[+] [PATTERN]</code>		list user mappings

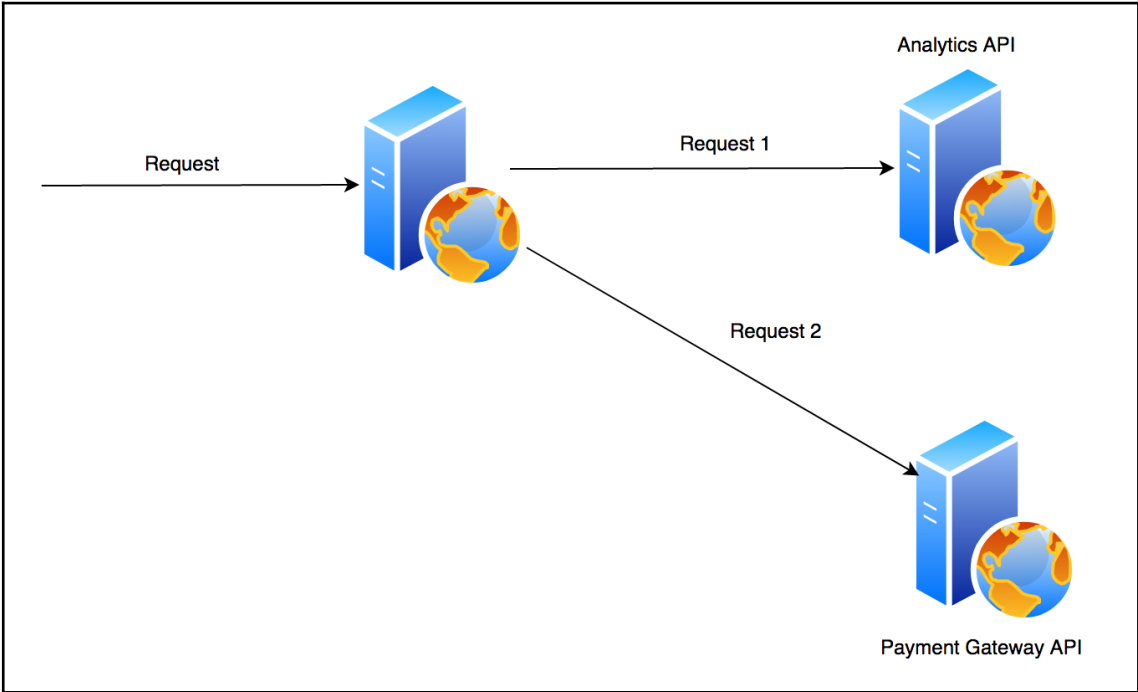
Chapter 8: Building a REST API Client in Go

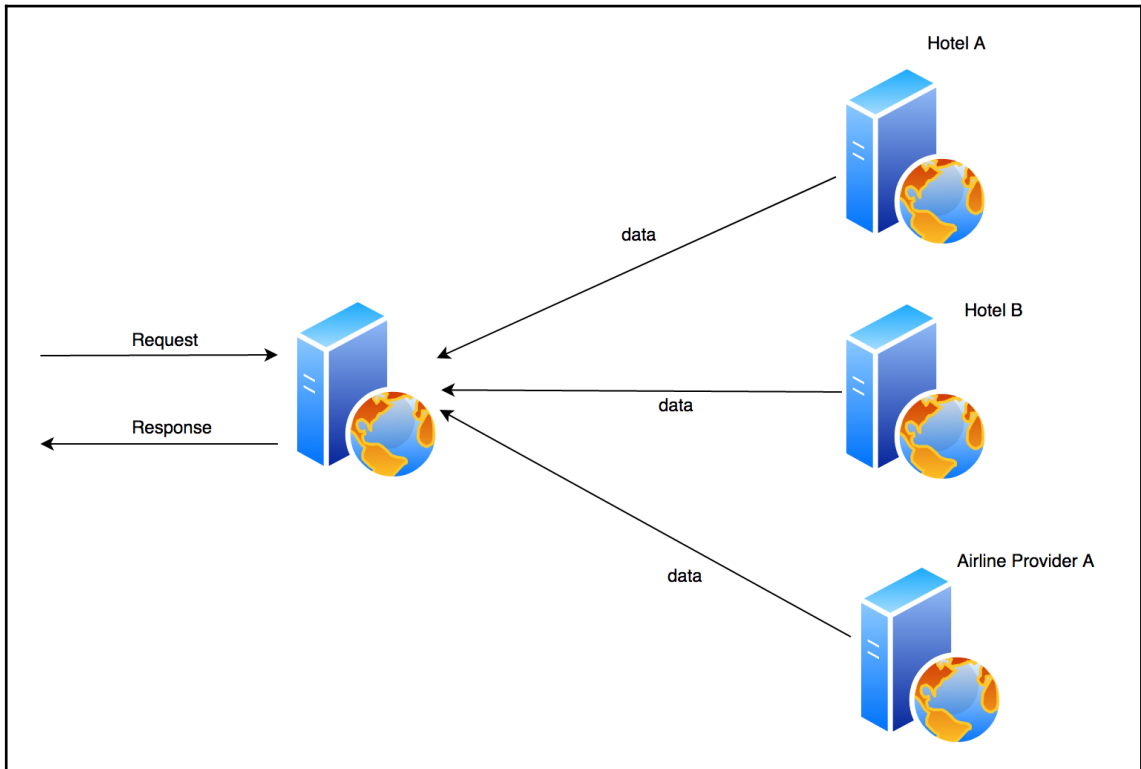


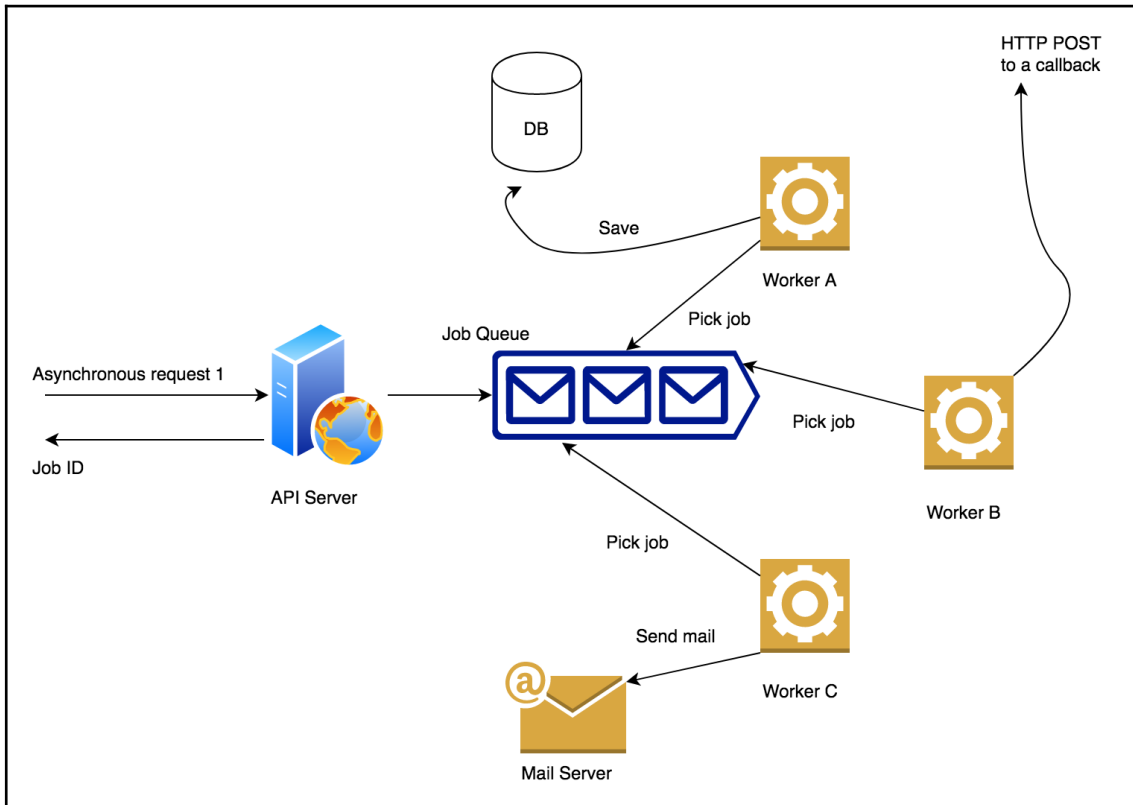


Chapter 9: Asynchronous API Design

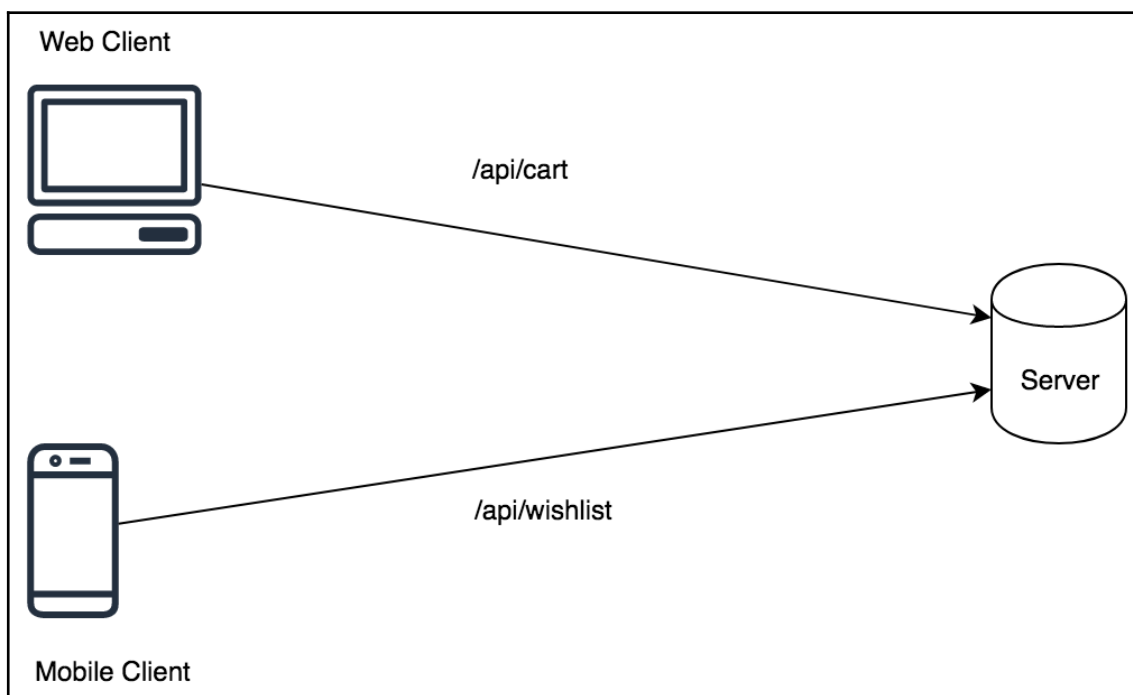


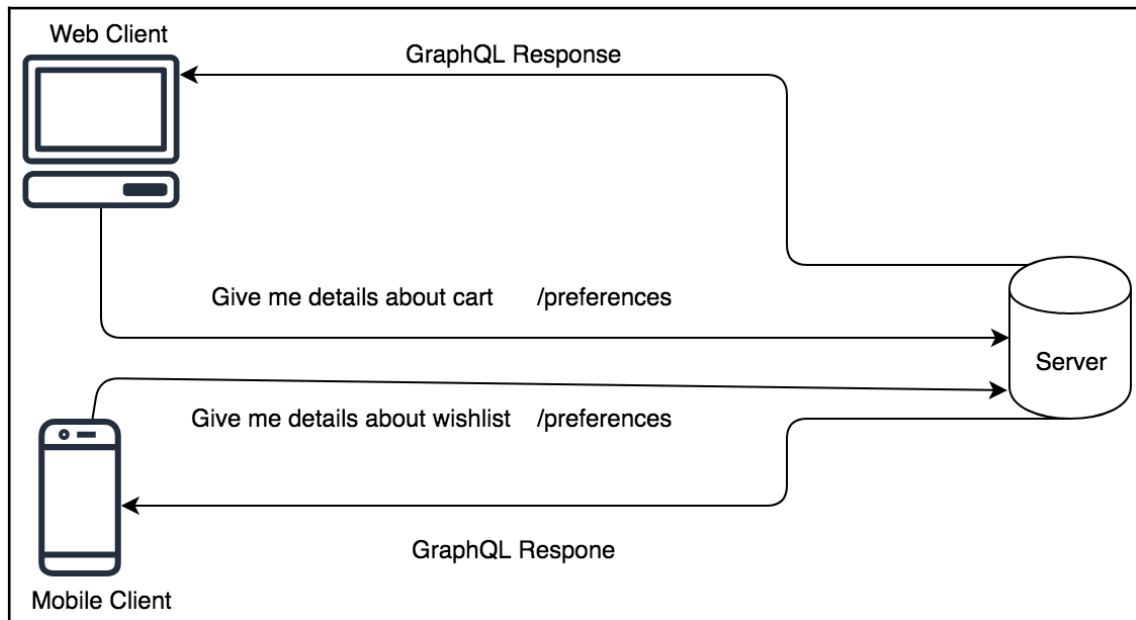




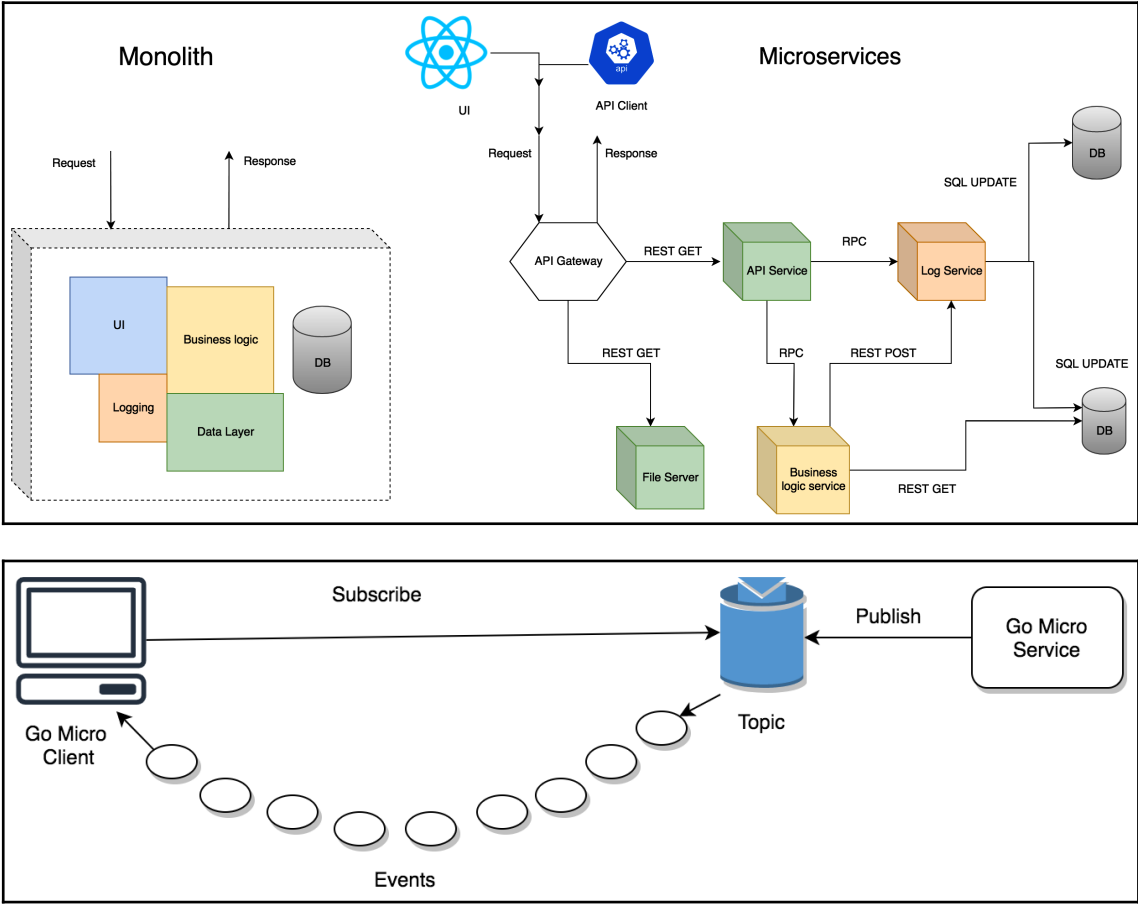


Chapter 10: GraphQL and Go





Chapter 11: Scaling our REST API Using Microservices



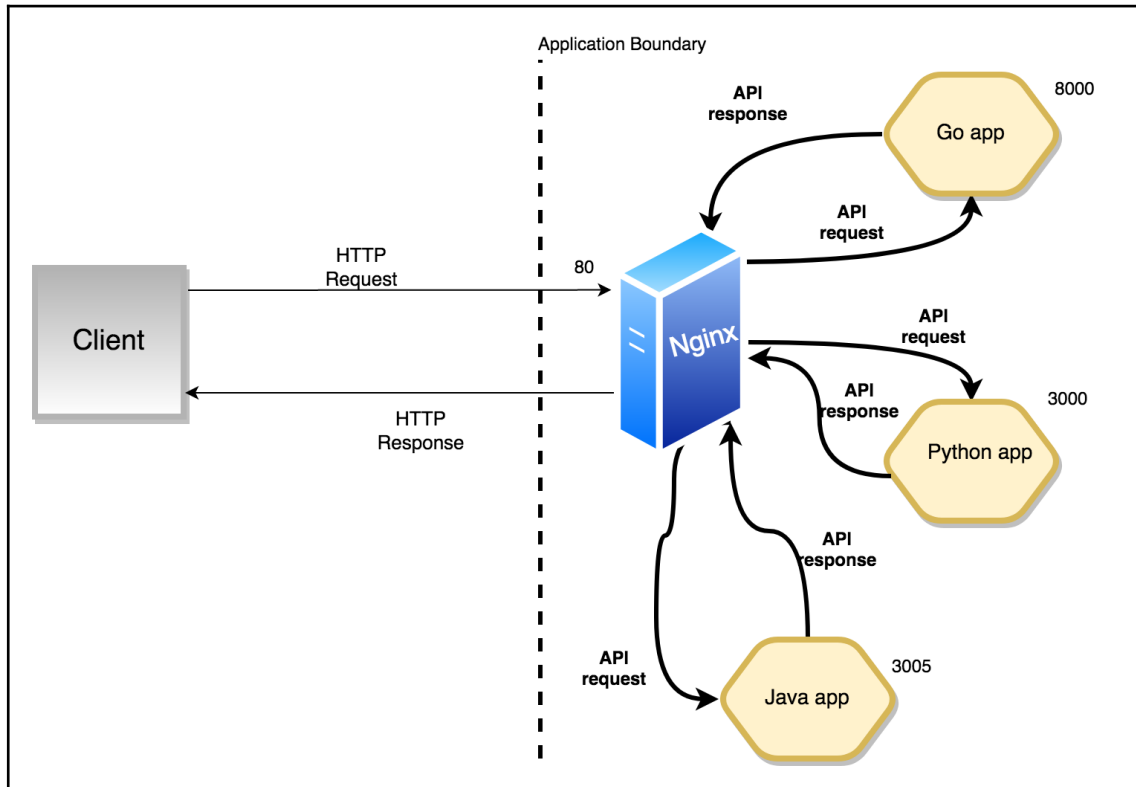
Chapter 12: Containerizing REST Services for Deployment

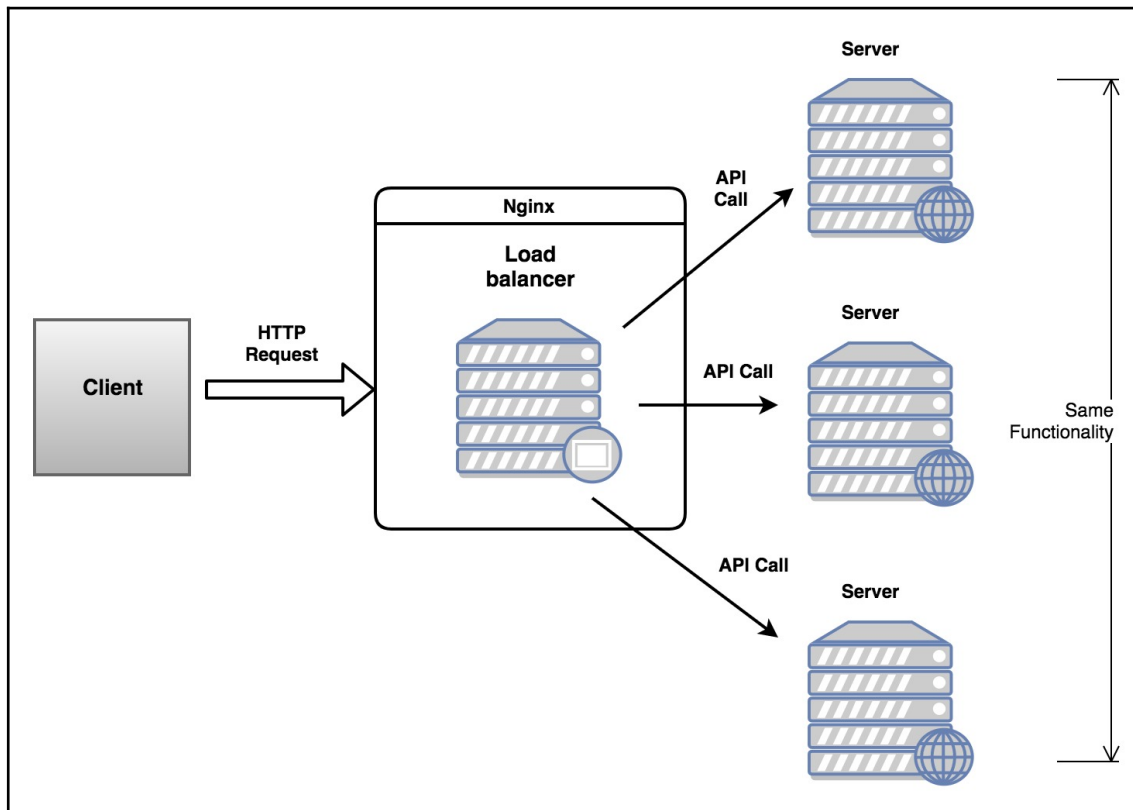
Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.





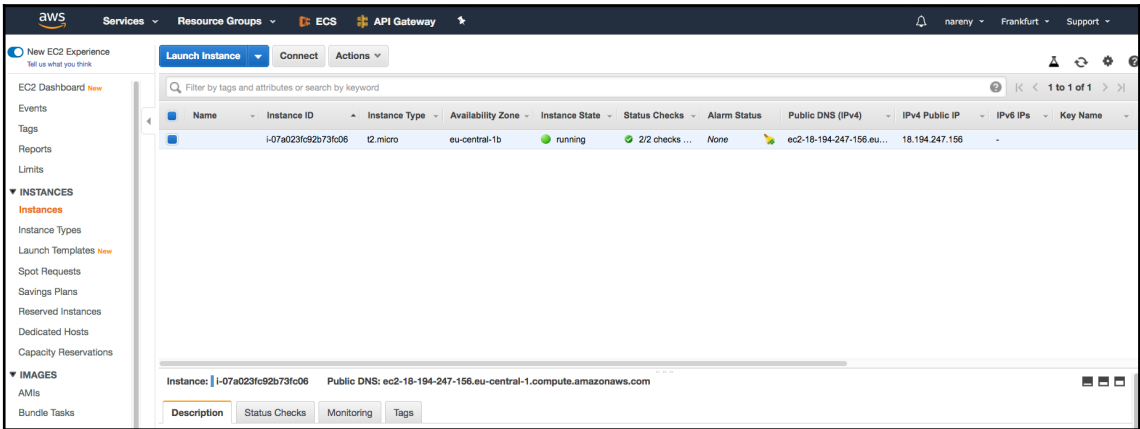
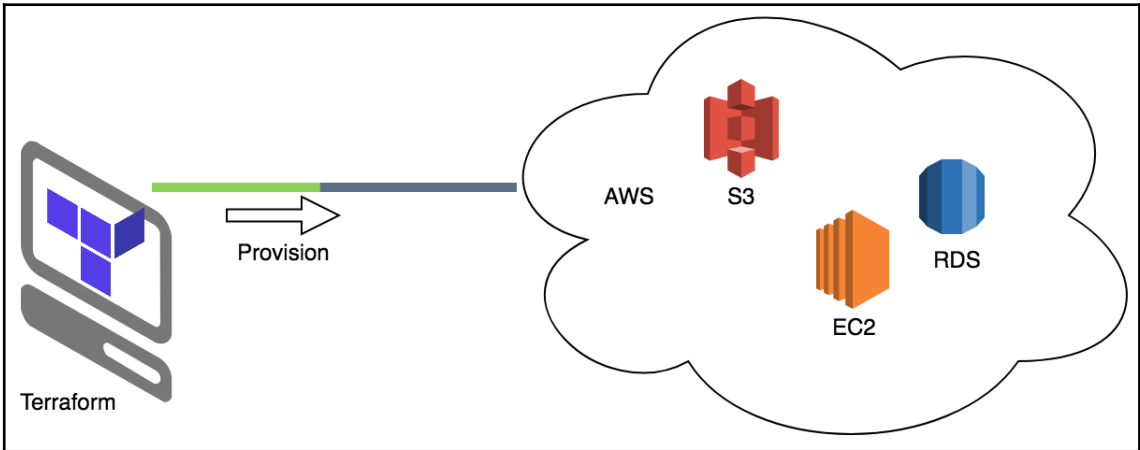
```
root@ubuntu:~# supervisorctl
myserver                                RUNNING    pid 6886, uptime 0:00:47
supervisor> help

default commands (type help <topic>):
=====
add      clear  fg      open  quit   remove restart start  stop  update
avail   exit   maintail pid  reload reread shutdown status tail  version
supervisor> █
```

```
root@ubuntu:~# kill 6886
root@ubuntu:~# supervisorctl
myserver                                RUNNING    pid 6903, uptime 0:00:10
supervisor> help

default commands (type help <topic>):
```

Chapter 13: Deploying REST Services on Amazon Web Services



aws

Services

Resource Groups

ECS

API Gateway

naren

Frankfurt

Support

New EC2 Experience

Tell us what you think

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

IMAGES

AMIs

Bundle Tasks

ELASTIC BLOCK STORE

Volumes

Snapshots

Lifecycle Manager

NETWORK & SECURITY

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Create Key Pair

Import Key Pair

Delete

Filter by attributes or search by keyword

1 to 1 of 1

Key pair name	Fingerprint
api-server-key	04:ee:79:e6:7a:95:a6:52:38:f1:ab:1d:33:aa:ae:c4

Key Pair: api-server-key

Key pair name

api-server-key

Fingerprint

04:ee:79:e6:7a:95:a6:52:38:f1:ab:1d:33:aa:ae:c4

[28]

aws

Services

Resource Groups

ECS

API Gateway

New EC2 Experience

Tell us what you think

Launch templates

Spot Requests

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Reserved Instances

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Capacity Reservations

IMAGES

AMIs

Bundle Tasks

ELASTIC BLOCK STORE

Volumes

Snapshots

Lifecycle Manager

NETWORK & SECURITY

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

LOAD BALANCING

Load Balancers

Target Groups

AUTO SCALING

Launch Configurations

Auto Scaling Groups

Launch Instance

Connect

Actions

Filter by tags and attributes or search by keyword

< > 1 to 1 of 1 >

Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (IPv4)	IPv4 Public IP	IPv6 IPs	Key Name
	i-050d7ec98b4d6a814	t2.micro	eu-central-1b	running	2/2 checks ...	None	ec2-52-59-192-138.eu-...	52.59.192.138	-	api-server-key

Instance: i-050d7ec98b4d6a814

Public DNS: ec2-52-59-192-138.eu-central-1.compute.amazonaws.com

Description

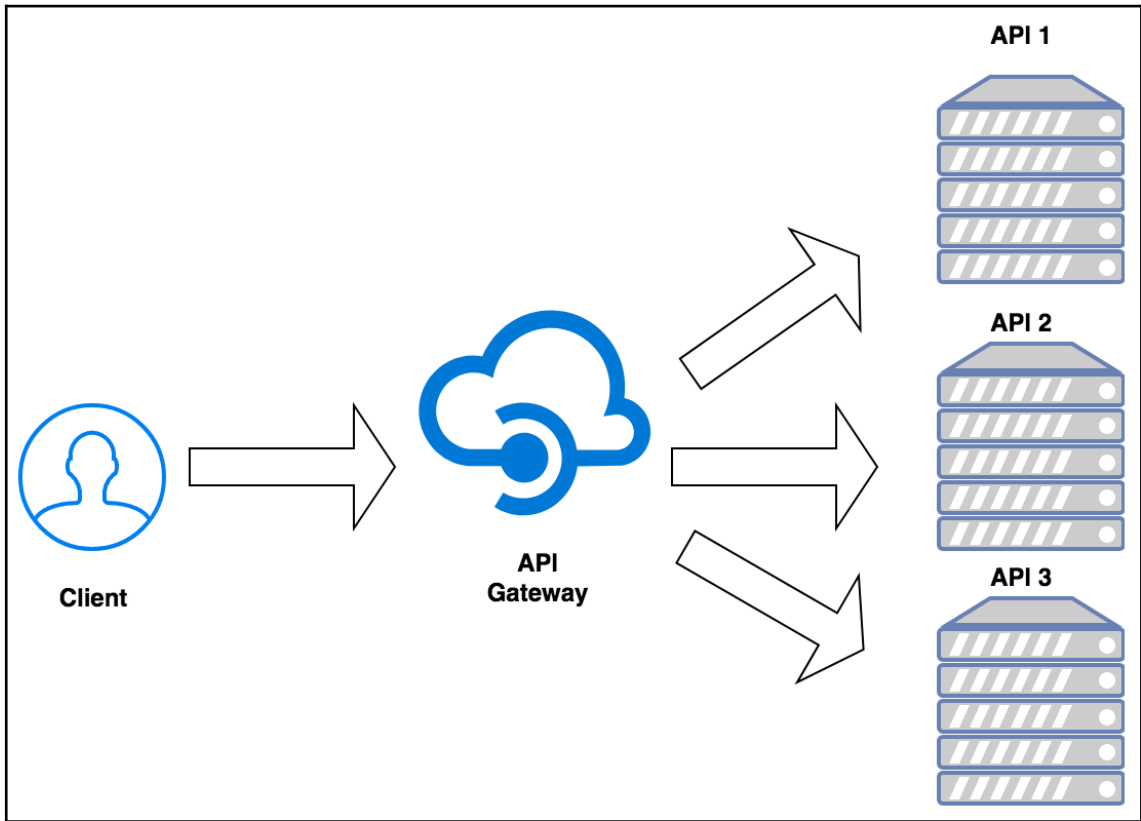
Status Checks

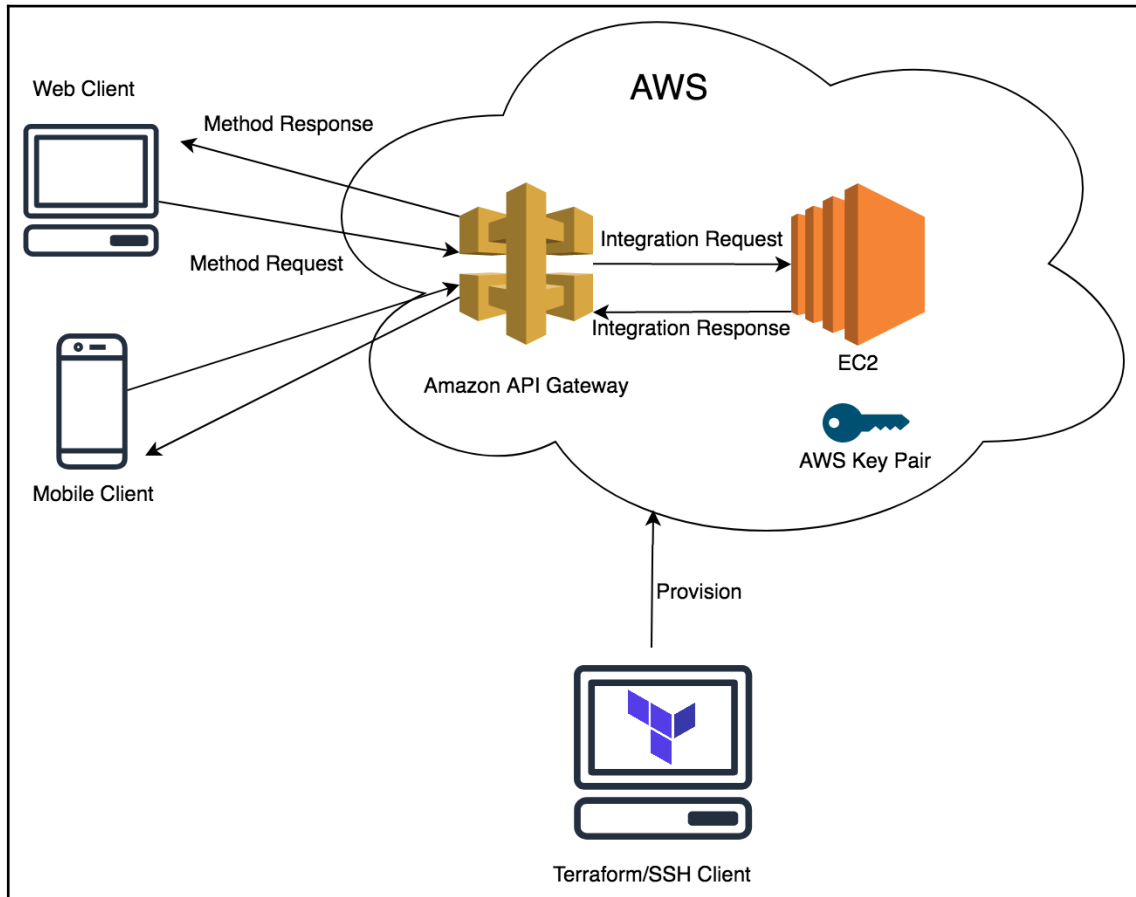
Monitoring

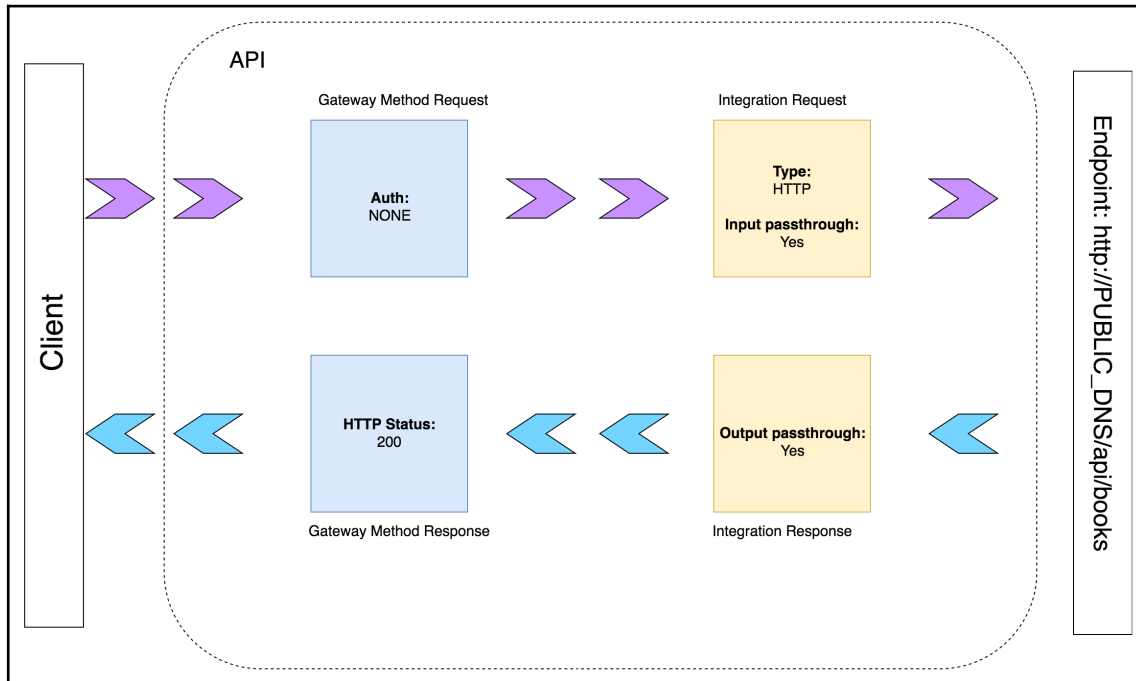
Tags

Instance ID	i-050d7ec98b4d6a814	Public DNS (IPv4)	ec2-52-59-192-138.eu-central-1.compute.amazonaws.com
Instance state	running	IPv4 Public IP	52.59.192.138
Instance type	t2.micro	IPv6 IPs	-
Elastic IPs		Private DNS	ip-172-31-37-219.eu-central-1.compute.internal
Availability zone	eu-central-1b	Private IPs	172.31.37.219
Security groups	default, view inbound rules, view outbound rules	Secondary private IPs	
Scheduled events	No scheduled events	VPC ID	vpc-fec31794
AMI ID	ami-ubuntu-18.04-1.16.2-00-1571834140 (ami-03816140b4ac9ae2b)	Subnet ID	subnet-504a502d
Platform	-	Network interfaces	eth0
IAM role	-	Source/dest. check	True
Key pair name	api-server-key	T2/T3 Unlimited	Disabled

Public IP







Chapter 14: Handling Authentication for our REST Services

